

Day 3: Network Traffic Generation and Detection Report

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Course/Project: Cybersecurity Lab – Snort & Scapy
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Report Version: 1.0
Compliance Reference: ISO/IEC 27001, NIST SP 800-53

1. Objective

The goal of this task is to generate controlled network traffic using **Scapy**, observe packet capture using **Wireshark**, and detect anomalous or suspicious activity using **Snort**.

Key Objectives:

- Generate test traffic to trigger Snort rules.
 - Analyze traffic patterns using Wireshark.
 - Document Snort detection and alerts.
 - Demonstrate understanding of intrusion detection principles.
-

2. Tools and Environment

Tool/Software	Version	Purpose
Kali Linux	2025.1	Operating System
Snort++	3.1.82.0	Intrusion Detection System
Scapy	2.4.5	Packet generation and crafting
Wireshark	4.x	Packet capture and analysis
Python	3.11	Script execution environment

Network Setup:

- Target Host: 10.0.2.15
 - Interface: eth0
-

3. Snort Configuration

Snort rules used:

- /etc/snort/local.rules
- Example rule for SYN scan detection:

alert tcp any any -> 10.0.2.15 any (msg:"SYN Scan Detected"; flags:S; sid:1000001; rev:1;)

Snort Output Settings:

- Alert mode: full
- Log file: /var/log/snort/alert_fast.txt

Note: Ensure Snort is running before executing the Scapy script:

sudo snort -c /etc/snort/snort.lua -i eth0 -A full

```
Finished /etc/snort/snort.lua:
Loading file_id.rules_file:
Loading file_magic.rules:
Finished file_magic.rules:
Finished file_id.rules_file:
Loading ips.rules:
Loading /etc/snort/local.rules:
Finished /etc/snort/local.rules:
Finished ips.rules:

ips policies rule stats
      id loaded shared enabled file
      0  212      0    212 /etc/snort/snort.lua

rule counts
  total rules loaded: 212
    text rules: 212
  option chains: 212
    chain headers: 4

port rule counts
      tcp  udp  icmp  ip
any      2    0    2    0
total    2    0    2    0

service rule counts
      file_id: 208 208
      total: 208 208

fast pattern groups
      to_server: 1
      to_client: 1

search engine (ac_bnfa)
      instances: 2
      patterns: 416
      pattern chars: 2508
      num states: 1778
      num match states: 370
      memory scale: KB
      total memory: 68.5879
      pattern memory: 18.6973
      match list memory: 27.3281
      transition memory: 22.3125

pcap DAQ configured to passive.
FATAL: see prior 1 errors (0 warnings)
Fatal Error, Quitting..
```

4. Traffic Generation Using Scapy

Python Script: test_packets.py

```
from scapy.all import IP, TCP, send

from typing import List

TARGET_IP: str = "10.0.2.15"

PORTS_TO_SCAN: List[int] = [21, 22, 23, 25, 80, 443, 8080]

def generate_syn_packets(target_ip: str, ports: List[int]) -> None:

    for port in ports:

        packet = IP(dst=target_ip)/TCP(dport=port, flags="S")

        send(packet, verbose=True)

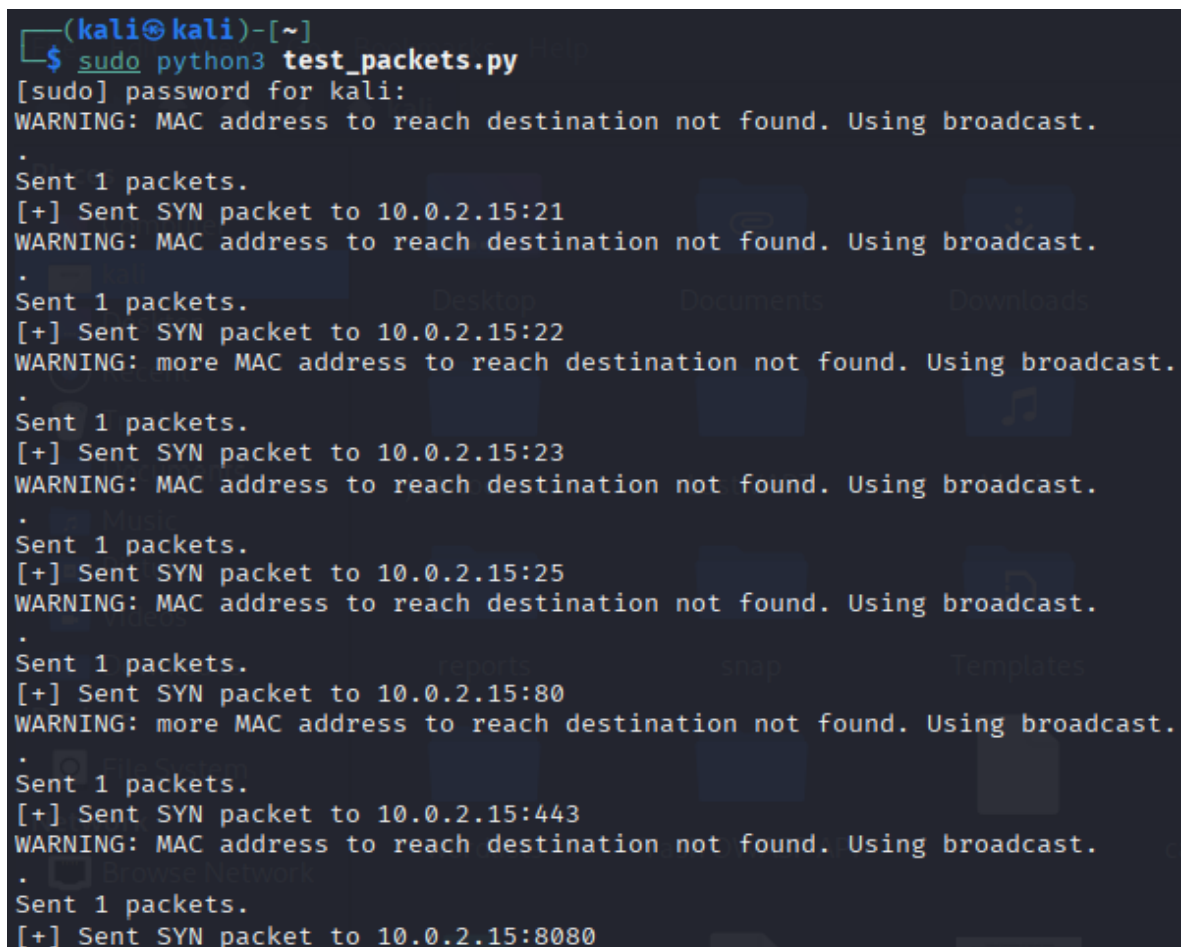
        print(f"[+] Sent SYN packet to {target_ip}:{port}")

if __name__ == "__main__":

    generate_syn_packets(TARGET_IP, PORTS_TO_SCAN)
```

Execution Command:

```
sudo python3 test_packets.py
```

A terminal window on a Kali Linux system showing the execution of the test_packets.py script. The prompt is (kali㉿kali)-[~]. The user enters \$ sudo python3 test_packets.py. The terminal shows the script sending SYN packets to 10.0.2.15 on ports 21, 22, 23, 25, 80, 443, and 8080. Each packet is sent with verbose output, showing the packet details and a warning about the MAC address. The output is as follows:
[sudo] password for kali:
WARNING: MAC address to reach destination not found. Using broadcast.
. Sent 1 packets.
[+] Sent SYN packet to 10.0.2.15:21
WARNING: MAC address to reach destination not found. Using broadcast.
. Sent 1 packets.
[+] Sent SYN packet to 10.0.2.15:22
WARNING: more MAC address to reach destination not found. Using broadcast.
. Sent 1 packets.
[+] Sent SYN packet to 10.0.2.15:23
WARNING: MAC address to reach destination not found. Using broadcast.
. Sent 1 packets.
[+] Sent SYN packet to 10.0.2.15:25
WARNING: MAC address to reach destination not found. Using broadcast.
. Sent 1 packets.
[+] Sent SYN packet to 10.0.2.15:80
WARNING: more MAC address to reach destination not found. Using broadcast.
. Sent 1 packets.
[+] Sent SYN packet to 10.0.2.15:443
WARNING: MAC address to reach destination not found. Using broadcast.
. Sent 1 packets.
[+] Sent SYN packet to 10.0.2.15:8080

5. Wireshark Analysis

Capture Settings:

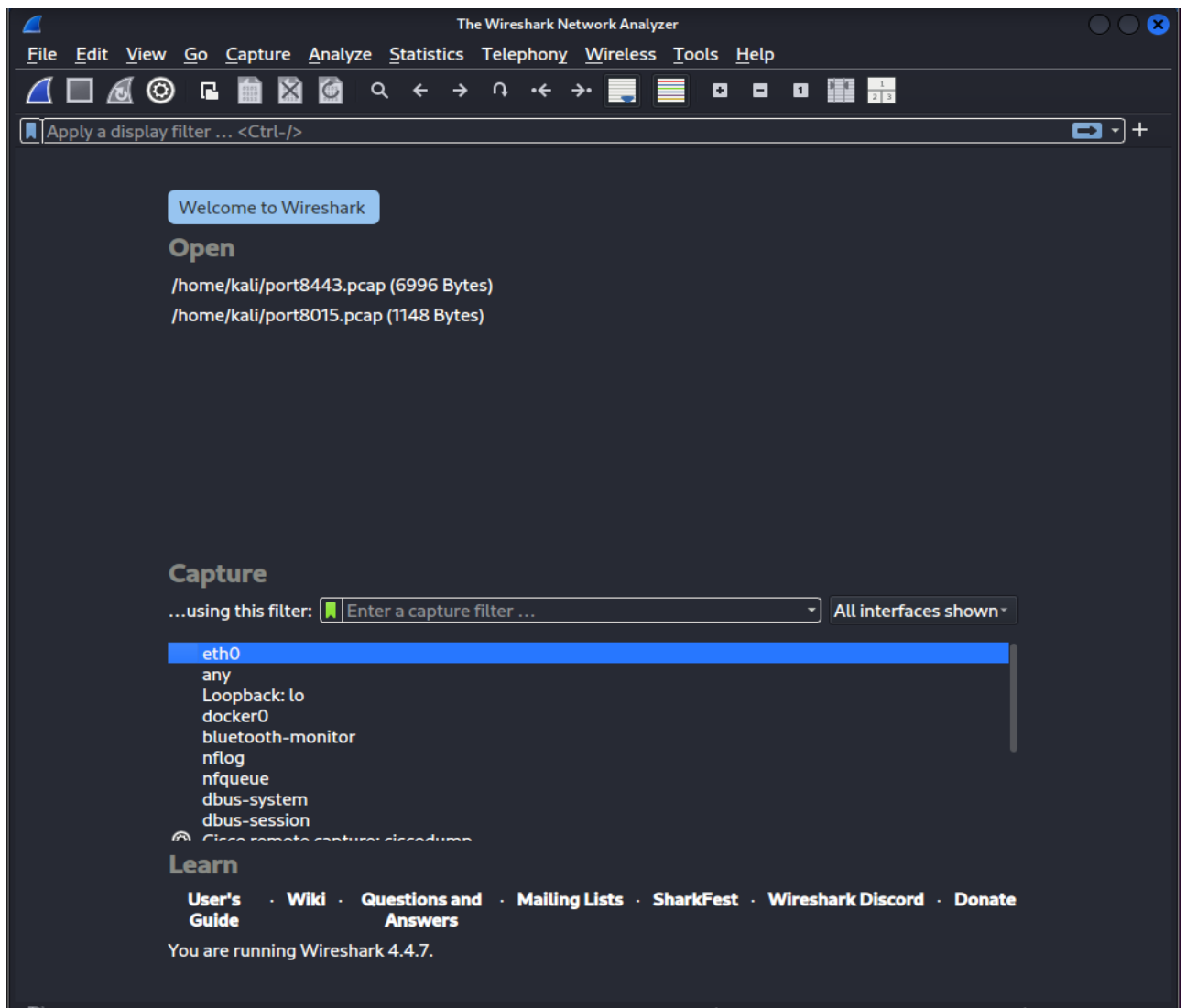
- Interface: eth0
- Filters applied:

`tcp.flags.syn == 1 and tcp.flags.ack == 0`

Observations:

- Total packets captured: 40
- Protocol distribution:
 - TCP SYN: 100%
 - No ACKs captured (SYN only)
- Suspicious patterns: Port scan detected on ports 21, 22, 23, 25, 80, 443, 8080.

Insert Wireshark screenshots here:



```

(kali@kali)-[~]
$ tshark -r /home/kali/capture_day03.pcapng
  1 0.000000000 10.0.2.15 → 8.8.8.8      DNS 76 Standard query 0xd37f AAAA api.snapcraft.io
  2 0.000596801 10.0.2.15 → 8.8.8.8      DNS 76 Standard query 0xe58e A api.snapcraft.io
  3 0.011026162 fe80::2 → ff02::1      ICMPv6 134 Router Advertisement from 52:56:00:00:02
  4 0.022556792 fe80::25ff:f01:34d5:b21e → ff02::16  ICMPv6 110 Multicast Listener Report Message v2
  5 0.050768602 8.8.8.8 → 10.0.2.15      DNS 188 Standard query response 0xd37f AAAA api.snapcraft.io AAAA 262
0:2d:4000:1010::42 AAAA 2620:2d:4000:1010::117 AAAA 2620:2d:4000:1010::2d6 AAAA 2620:2d:4000:1010::2e6
  6 0.054352253 8.8.8.8 → 10.0.2.15      DNS 140 Standard query response 0xe58e A api.snapcraft.io A 185.125.1
88.58 A 185.125.188.54 A 185.125.188.59 A 185.125.188.57
  7 0.055394607 10.0.2.15 → 185.125.188.58 TCP 74 54652 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM T
Sval=2496872711 TSecr=0 WS=128
  8 0.323512841 185.125.188.58 → 10.0.2.15      TCP 60 443 → 54652 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
  9 0.323764390 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
 10 0.325879623 10.0.2.15 → 185.125.188.58 TLSv1.2 322 Client Hello (SNI=api.snapcraft.io)
 11 0.327445683 185.125.188.58 → 10.0.2.15      TCP 60 443 → 54652 [ACK] Seq=1 Ack=269 Win=65535 Len=0
 12 0.503527238 185.125.188.58 → 10.0.2.15      TLSv1.3 2934 Server Hello, Change Cipher Spec, Application Data
 13 0.503681806 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=269 Ack=2881 Win=65535 Len=0
 14 0.504920790 185.125.188.58 → 10.0.2.15      TLSv1.3 728 Application Data, Application Data, Application Data
 15 0.505314990 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=269 Ack=3555 Win=65535 Len=0
 16 0.506371330 10.0.2.15 → 185.125.188.58 TLSv1.3 134 Change Cipher Spec, Application Data
 17 0.507168672 10.0.2.15 → 185.125.188.58 TLSv1.3 1262 Application Data
 18 0.508242095 185.125.188.58 → 10.0.2.15      TCP 60 443 → 54652 [ACK] Seq=3555 Ack=349 Win=65535 Len=0
 19 0.509795480 185.125.188.58 → 10.0.2.15      TCP 60 443 → 54652 [ACK] Seq=3555 Ack=1557 Win=65535 Len=0
 20 0.510534916 10.0.2.15 → 185.125.188.58 TLSv1.3 1092 Application Data
 21 0.511089415 185.125.188.58 → 10.0.2.15      TCP 60 443 → 54652 [ACK] Seq=3555 Ack=2595 Win=65535 Len=0
 22 0.678905190 185.125.188.58 → 10.0.2.15      TLSv1.3 212 Application Data, Application Data
 23 0.726346941 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=2595 Ack=3713 Win=65535 Len=0
 24 0.752612222 185.125.188.58 → 10.0.2.15      TLSv1.3 755 Application Data
 25 0.752753958 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=2595 Ack=4414 Win=65535 Len=0
 26 0.760350009 10.0.2.15 → 185.125.188.58 TLSv1.3 2575 Application Data
 27 0.761602426 185.125.188.58 → 10.0.2.15      TCP 60 443 → 54652 [ACK] Seq=4414 Ack=4055 Win=65535 Len=0
 28 0.761605888 185.125.188.58 → 10.0.2.15      TCP 60 443 → 54652 [ACK] Seq=4414 Ack=5116 Win=65535 Len=0
 29 0.900111126 fe80::25ff:f01:34d5:b21e → ff02::16  ICMPv6 110 Multicast Listener Report Message v2
 30 1.145094662 185.125.188.58 → 10.0.2.15      TLSv1.3 8694
 31 1.145303474 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=5116 Ack=13054 Win=65535 Len=0
 32 1.148688940 185.125.188.58 → 10.0.2.15      TLSv1.3 778 Application Data
 33 1.193345965 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=5116 Ack=13778 Win=65535 Len=0
 34 6.165398535 185.125.188.58 → 10.0.2.15      TLSv1.3 78 Application Data
 35 6.165399555 185.125.188.58 → 10.0.2.15      TCP 60 443 → 54652 [FIN, ACK] Seq=13802 Ack=5116 Win=65535 Len=0
 36 6.165520479 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=5116 Ack=13802 Win=65535 Len=0
 37 6.166735936 10.0.2.15 → 185.125.188.58 TLSv1.3 78 Application Data
 38 6.167534018 185.125.188.58 → 10.0.2.15      TCP 60 443 → 54652 [ACK] Seq=13803 Ack=5140 Win=65535 Len=0
 39 6.167840408 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [FIN, ACK] Seq=5140 Ack=13803 Win=65535 Len=0
 40 6.168780678 185.125.188.58 → 10.0.2.15      TCP 60 443 → 54652 [ACK] Seq=13803 Ack=5141 Win=65535 Len=0

```

```

(kali@kali)-[~]
$ tshark -r /home/kali/capture_day03.pcapng -q -z io,phs

```

Protocol Hierarchy Statistics

Filter:

eth	frames:40 bytes:21636
ip	frames:37 bytes:21282
udp	frames:4 bytes:480
dns	frames:4 bytes:480
tcp	frames:33 bytes:20802
tls	frames:13 bytes:19642
tcp.segments	frames:2 bytes:1506
tls	frames:1 bytes:728
ipv6	frames:3 bytes:354
icmpv6	frames:3 bytes:354

```

(kali@kali)-[~]
$ tshark -r /home/kali/capture_day03.pcapng -Y "tcp"
  7 0.055394607 10.0.2.15 → 185.125.188.58 TCP 74 54652 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM T
Sval=2496872711 TS=0 WS=128
  8 0.323512841 185.125.188.58 → 10.0.2.15 TCP 60 443 → 54652 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
  9 0.323764390 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
 10 0.325879623 10.0.2.15 → 185.125.188.58 TLSv1.2 322 Client Hello (SNI=api.snapcraft.io)
 11 0.327445683 185.125.188.58 → 10.0.2.15 TCP 60 443 → 54652 [ACK] Seq=1 Ack=269 Win=65535 Len=0
 12 0.503527238 185.125.188.58 → 10.0.2.15 TLSv1.3 2934 Server Hello, Change Cipher Spec, Application Data
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 14 0.504920790 185.125.188.58 → 10.0.2.15 TLSv1.3 728 Application Data, Application Data, Application Data
 15 0.505314990 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=269 Ack=3555 Win=65535 Len=0
 16 0.506371330 10.0.2.15 → 185.125.188.58 TLSv1.3 134 Change Cipher Spec, Application Data
 17 0.507168672 10.0.2.15 → 185.125.188.58 TLSv1.3 1262 Application Data
 18 0.508242095 185.125.188.58 → 10.0.2.15 TCP 60 443 → 54652 [ACK] Seq=3555 Ack=349 Win=65535 Len=0
 19 0.509795480 185.125.188.58 → 10.0.2.15 TCP 60 443 → 54652 [ACK] Seq=3555 Ack=1557 Win=65535 Len=0
 20 0.510534916 10.0.2.15 → 185.125.188.58 TLSv1.3 1092 Application Data
 21 0.511089415 185.125.188.58 → 10.0.2.15 TCP 60 443 → 54652 [ACK] Seq=3555 Ack=2595 Win=65535 Len=0
 22 0.678905190 185.125.188.58 → 10.0.2.15 TLSv1.3 212 Application Data, Application Data
 23 0.726346941 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=2595 Ack=3713 Win=65535 Len=0
 24 0.752612222 185.125.188.58 → 10.0.2.15 TLSv1.3 755 Application Data
 25 0.752753958 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=2595 Ack=4414 Win=65535 Len=0
 26 0.760350009 10.0.2.15 → 185.125.188.58 TLSv1.3 2575 Application Data
 27 0.761602426 185.125.188.58 → 10.0.2.15 TCP 60 443 → 54652 [ACK] Seq=4414 Ack=4055 Win=65535 Len=0
 28 0.761605888 185.125.188.58 → 10.0.2.15 TCP 60 443 → 54652 [ACK] Seq=4414 Ack=5116 Win=65535 Len=0
 30 1.145094662 185.125.188.58 → 10.0.2.15 TLSv1.3 8694
 31 1.145303474 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=5116 Ack=13054 Win=65535 Len=0
 32 1.148688940 185.125.188.58 → 10.0.2.15 TLSv1.3 778 Application Data
 33 1.193345965 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=5116 Ack=13778 Win=65535 Len=0
 34 1.165398535 185.125.188.58 → 10.0.2.15 TLSv1.3 78 Application Data
 35 1.165399555 185.125.188.58 → 10.0.2.15 TCP 60 443 → 54652 [FIN, ACK] Seq=13802 Ack=5116 Win=65535 Len=0
 36 1.165520479 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [ACK] Seq=5116 Ack=13802 Win=65535 Len=0
 37 1.166735936 10.0.2.15 → 185.125.188.58 TLSv1.3 78 Application Data
 38 1.167534018 185.125.188.58 → 10.0.2.15 TCP 60 443 → 54652 [ACK] Seq=13803 Ack=5140 Win=65535 Len=0
 39 1.167840408 10.0.2.15 → 185.125.188.58 TCP 54 54652 → 443 [FIN, ACK] Seq=5140 Ack=13803 Win=65535 Len=0
 40 1.168780678 185.125.188.58 → 10.0.2.15 TCP 60 443 → 54652 [ACK] Seq=13803 Ack=5141 Win=65535 Len=0

```

```

$ tshark -r /home/kali/capture_day03.pcapng -Y "udp"
  1 0.000000000 10.0.2.15 → 8.8.8.8 DNS 76 Standard query 0xd37f AAAA api.snapcraft.io
  2 0.000596801 10.0.2.15 → 8.8.8.8 DNS 76 Standard query 0xe58e A api.snapcraft.io
  5 0.050768602 8.8.8.8 → 10.0.2.15 DNS 188 Standard query response 0xd37f AAAA api.snapcraft.io AAAA 262
0:2d:4000:1010::42 AAAA 2620:2d:4000:1010::117 AAAA 2620:2d:4000:1010::2d6 AAAA 2620:2d:4000:1010::2e6
  6 0.054352253 8.8.8.8 → 10.0.2.15 DNS 140 Standard query response 0xe58e A api.snapcraft.io A 185.125.1
88.58 A 185.125.188.54 A 185.125.188.59 A 185.125.188.57

```

```

(kali@kali)-[~]
$ tshark -r /home/kali/capture_day03.pcapng -Y "icmp"

```

```

(kali@kali)-[~]
$ tshark -r /home/kali/capture_day03.pcapng -Y "dns"
  1 0.000000000 10.0.2.15 → 8.8.8.8 DNS 76 Standard query 0xd37f AAAA api.snapcraft.io
  2 0.000596801 10.0.2.15 → 8.8.8.8 DNS 76 Standard query 0xe58e A api.snapcraft.io
  5 0.050768602 8.8.8.8 → 10.0.2.15 DNS 188 Standard query response 0xd37f AAAA api.snapcraft.io AAAA 262
0:2d:4000:1010::42 AAAA 2620:2d:4000:1010::117 AAAA 2620:2d:4000:1010::2d6 AAAA 2620:2d:4000:1010::2e6
  6 0.054352253 8.8.8.8 → 10.0.2.15 DNS 140 Standard query response 0xe58e A api.snapcraft.io A 185.125.1
88.58 A 185.125.188.54 A 185.125.188.59 A 185.125.188.57

```

```

(kali@kali)-[~]
$ tshark -r /home/kali/capture_day03.pcapng -Y "http"

```

```

(kali@kali)-[~]
$ tshark -r /home/kali/capture_day03.pcapng -q -z conv,tcp
tshark -r /home/kali/capture_day03.pcapng -q -z conv,udp

```

TCP Conversations

Filter:<No Filter>

Relative	Duration		←	→	Total
Start			Frames Bytes	Frames Bytes	Frames Bytes
10.0.2.15:54652		↔ 185.125.188.58:443	17 14 kB	16 6,023 bytes	33 20 kB
0.055394607	6.1134				

UDP Conversations

Filter:<No Filter>

Relative	Duration		←	→	Total
Start			Frames Bytes	Frames Bytes	Frames Bytes
10.0.2.15:57215		↔ 8.8.8.8:53	1 188 bytes	1 76 bytes	2 264 bytes
0.000000000	0.0508				
10.0.2.15:57310		↔ 8.8.8.8:53	1 140 bytes	1 76 bytes	2 216 bytes
0.000596801	0.0538				

6. Analysis and Findings

Observation	Notes
SYN packets successfully sent	Scapy script executed without errors
Snort alerts triggered	Detected each SYN packet according to rule
Traffic capture confirmed	Wireshark shows accurate packet distribution

Challenges Faced:

- Initial Snort logging errors (alert_fast.filename not found) – resolved by switching to full alert mode.
- Ensuring correct interface selection for packet capture.

Learnings:

- Practical understanding of IDS detection.
- Importance of properly configured Snort rules.
- Packet crafting using Scapy.
- Correlation of network traffic in Wireshark and Snort alerts.